# Maurizio Ungaro

Biography

Staff Scientist, Jefferson Lab | Nuclear physicist and simulation software developer

ungaro@jlab.org maureeungaro.github.io/home github.com/maureeungaro

Google Scholar INSPIRE ORCID 0000-0001-6982-3310 LinkedIn

## Short Bio

Maurizio Ungaro is a Staff Scientist at Jefferson Lab working on Geant4 simulations, GEMC, CLAS12 software, detector systems, and nucleon-structure research.

## Medium Bio

Maurizio Ungaro is a Staff Scientist in Jefferson Lab Hall-B. His work spans nucleon-structure measurements, CLAS12 detector systems, Geant4-based simulation, GEMC development, and distributed simulation production workflows. He develops tools, tutorials, and technical documentation that help scientists build and run reproducible detector simulations.

## Long Bio

Maurizio Ungaro is a nuclear physicist and simulation software developer at Jefferson Lab. His research focuses on the internal structure and dynamics of the nucleon, including meson electro-production and the transition between hadronic and partonic degrees of freedom. His technical work connects detector operation, Geant4 simulation, CLAS12 production workflows, and scientific software infrastructure.

He develops and supports GEMC, a database-driven Geant4 simulation workflow, and maintains CLAS12 simulation releases and production workflows. He also works on the CLAS12 Low Threshold Cherenkov Counter, including operation, maintenance, calibration, and detector-performance studies. In addition, he provides Geant4 tutorials and support material for Jefferson Lab users.

His earlier and collaborative work includes experiments at SPring-8 and participation in the Heavy Photon Search collaboration.

## ORCID / LinkedIn Draft

**ORCID biography**: Staff Scientist at Jefferson Lab working on nuclear physics, Geant4 simulations, GEMC, CLAS12 software, detector systems, and scientific computing workflows.

**LinkedIn headline**: Nuclear Physicist | Geant4 and GEMC Simulation Developer | CLAS12 Software and Detector Systems | Jefferson Lab